

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Social Engineering Simulation

Questions

1. As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain **tesla.com** using **theHarvester** in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyse how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks, and submit a report with screenshots, commands executed, findings, risk analysis, and recommendations. **(20 Marks)**

Commands:

1. `theHarvester -d tesla.com -b all`
2. `theHarvester --version`

Purpose : This command performs an OSINT scan on the target domain tesla.com using the Rapid DNS search engine. It collects publicly available information such as email addresses, subdomains, hosts, and IP addresses.

Findings :

- **Emails:** 0 (No public emails found)
- **Hosts/Subdomains:** 668
- **IP Addresses:** 53
- **Interesting URLs:** 19

Risk Analysis

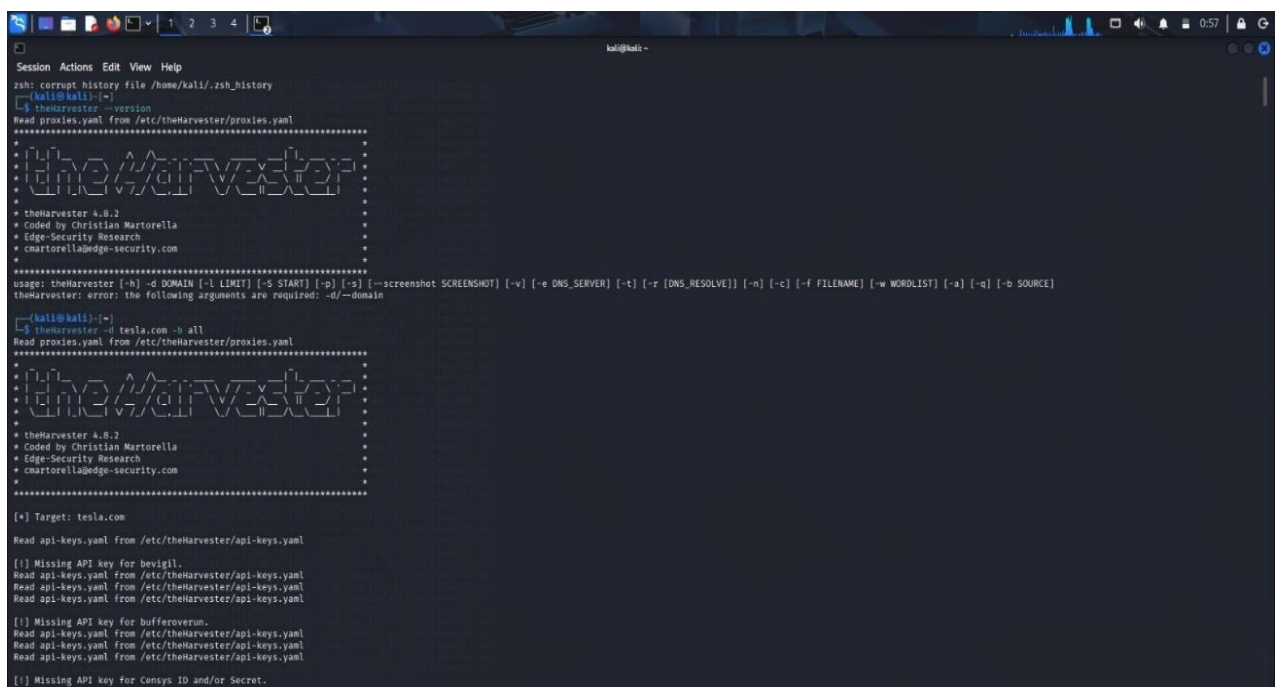
- Public hosts and URLs reveal Tesla's external infrastructure.
- Authentication portals may be targeted for credential harvesting.
- Public service names can be used in spear-phishing campaigns.
- Infrastructure details support reconnaissance and impersonation attacks.

Recommendations

1. Conduct regular OSINT assessments to identify exposed information.
2. Enable Multi-Factor Authentication (MFA).
3. Train employees to recognize phishing attacks.
4. Reduce unnecessary exposure of public subdomains.
5. Continuously monitor public-facing assets and DNS records.

Conclusion :

The OSINT assessment successfully identified public hosts/subdomains related to tesla.com. No public email addresses or IP addresses were discovered. The collected information demonstrates how attackers can gather infrastructure details during the reconnaissance phase and highlights the importance of minimizing public information exposure and implementing strong security controls.



```
kali@kali:~$ zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ theHarvester --version
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-g] [--screenshot SCREENSHOT] [-v] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain

kali@kali:~$ theHarvester -d tesla.com -h all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: tesla.com
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Consys ID and/or Secret.
```

```
Session Terminal Emulator Use the command line
[+] IPs found: 53
104.21.19.154
104.21.27.142
104.21.63.149
104.21.66.44
104.21.70.191
104.21.90.12
116.153.78.163
172.67.138.202
172.67.162.127
172.67.147.31
172.67.156.172
172.67.176.167
172.67.72.83
180.114.96.3
180.114.97.3
193.187.129.25
196.251.107.150
196.251.63.100
2.18.108.60
2.18.224.64
2.19.215.60
23.207.192.63
23.209.200.91
23.221.132.86
23.222.196.70
23.222.90.70
23.34.124.53
23.39.184.70
23.51.40.61
23.52.180.57
23.58.104.70
23.62.19.65
23.62.104.69
2080:1400:c400:1804::700
2080:1400:c400:180b::700
2080:1400:a800:791::700
2080:1400:a800:79c::700
2080:1700:3032::6015:3f95
2a02:26f0:1700:38b::700
2a02:26f0:1700:a::b056:6704
2a02:26f0:1700:b0d::700
2a02:26f0:2700:d93::700
2a02:26f0:3400:18c::700
2a02:26f0:3400:190::700
2a02:26f0:ab00:686::700
2a02:26f0:c300:38f::700
2a02:26f0:ed00:50b::700
2a06:98c1:3120::3
```

```
Session Terminal Emulator Use the command line
[+] No people found
[+] Hosts found: 668
13494342.auth.tesla.com
13494342u.origin-warp.tesla.com
2fdigitalassets.tesla.com
2fir.tesla.com
3.tesla.com
3a.model3.tesla.com
7offline-cloud-stage-uhsouthflorida.tesla.com
a.smf13.tcs.tesla.com
a659a85a888ef088124dc0e3d29c1.tesla.com
aa.ad.tesla.com
accounts.tesla.com:22.197.44.40
acme-sentry-aa.eng.usel.vn.cloud.digitalassets-shop.tesla.com
additional.tesla.com
adev.tesla.com
admin15.tesla.com
again.tesla.com
akamai-apigateway-automation-billing.origin-joule.tesla.com
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-bolt-forms.origin-finops.tesla.com
akamai-apigateway-bolt-forms.partners.tesla.com
akamai-apigateway-bolt-forms.tesla.com
akamai-apigateway-c360-services-prd.tesla.com
akamai-apigateway-c360-services-stg-npuv-eng.usel.vn.cloud.tesla.com
akamai-apigateway-c360-services-stg.tesla.com
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-charging-ownership-stage.sip.tesla.com
akamai-apigateway-claims-integration-api-stg-npuv-eng.usel.vn.cloud.tesla.com
akamai-apigateway-clamp-stg-dataviz-one.tesla.com
akamai-apigateway-crawford-stgm.auth.tesla.com
akamai-apigateway-dev-warptmsapiserver-upgrade.origin-joule.tesla.com
akamai-apigateway-dev-warptmsapiserver-upgrade.tesla.com
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-dev-storage-api.akamai-apigateway-translationpipeline.tesla.com
akamai-apigateway-ehs-stg-de-dataviz-one.tesla.com
akamai-apigateway-ehs.de.tesla.com
akamai-apigateway-ehs.stg.tesla.com
akamai-apigateway-fimplat-stg.origin-vmanage-alerts.tesla.com
akamai-apigateway-fimplateng-defa.partners.tesla.com
akamai-apigateway-fta-nrsproy00.tesla.com
akamai-apigateway-fta.tesla.com:akamai-apigateway-fta.tesla.com.edgekey.net
akamai-apigateway-gallagher-external-api.tesla.com
akamai-apigateway-l360-services-prd.model3.tesla.com
akamai-apigateway-l360-services-stg-cdn-design.tesla.com
akamai-apigateway-logisticratesapi.dataviz-one.tesla.com
akamai-apigateway-logisticratesapi.tesla.com:akamai-apigateway-logisticratesapi.tesla.com.edgekey.net
akamai-apigateway-logisticsignals.partners.tesla.com
akamai-apigateway-materials.tesla.com
akamai-apigateway-efs-stg-de-nrsproy00.tesla.com
```

```
Session Actions Edit View Help
ownerapi-alpha.prd.vn.cloud.tesla.com:66.17.6.26
ownerapi-api-eng.euw1.vn.cloud.tesla.com:52.50.13.50
ownerapi-api-eng.usel.vn.cloud.tesla.com:66.17.6.26
ownerapi-api.prd.usw.vn.cloud.tesla.com:66.17.8.14
pages.email.tesla.com:pages.virt.s7.exacttarget.com
partners.tesla.com
pc-family.tesla.com
philexton12robot.tesla.com
pm000.tesla.com
powerhub.energy.origin-joule.tesla.com
powerhub.energy.tesla.com:powerhub.se.tesla.services
prd-na.vn.cloud.os.tesla.com
professional.tesla.com
ptr1.tesla.com
ptr1.tesla.com:117.50.35.199
ptr2.tesla.com:117.50.16.178
raw.tesla.com
reply.github.tesla.com
r121ca.tesla.com
s3.a.iad006.tcs.tesla.com
s3.a.smf13.tcs.tesla.com
s3.eng-na.vn.cloud.tesla.com:66.17.8.21
s3.prd-na.vn.cloud.tesla.com:66.17.6.65
sales-prd.tesla.com:external-na-pop1.edgekey.net
samlbackchannel.tesla.com
schedule1.tesla.com
sentry-dev.tesla.com
sentry-api.ops-na.vn.cloud.tesla.com:66.17.8.100
sentry-app.tesla.com
service-npuv-eng.usel.vn.cloud.tesla.com
service.origin-vmanage-alerts.tesla.com
service.tesla.com
shop.tesla.com
signaling-robotics.eng.usw.vn.cloud.tesla.com:66.17.8.53
signaling-robotics.eng.vn.cloud.tesla.com:66.17.6.53
sso-dev.tesla.com
stage.tesla.com
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-event-dev.tesla.com
static-event-dev.tesla.com
static-pay-origin-mwcdn.tesla.com:209.133.79.52
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com
tcs-ga.tesla.com
tcs.tesla.com
teamchat.tesla.com:teamchat.tesla.com.edgekey.net
tesla-hermes-snapshot-motors-eng.s3.eng-na.vn.cloud.tesla.com:66.17.6.21
tesla-hermes-snapshot-motors-eng.usw.vn.cloud.tesla.com:66.17.6.54
teslaimgapi.tesla.com
teslatequila.tesla.com
tomato.tesla.com
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
```

2. As a cybersecurity awareness consultant, you are required to conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks.(20 Mark).

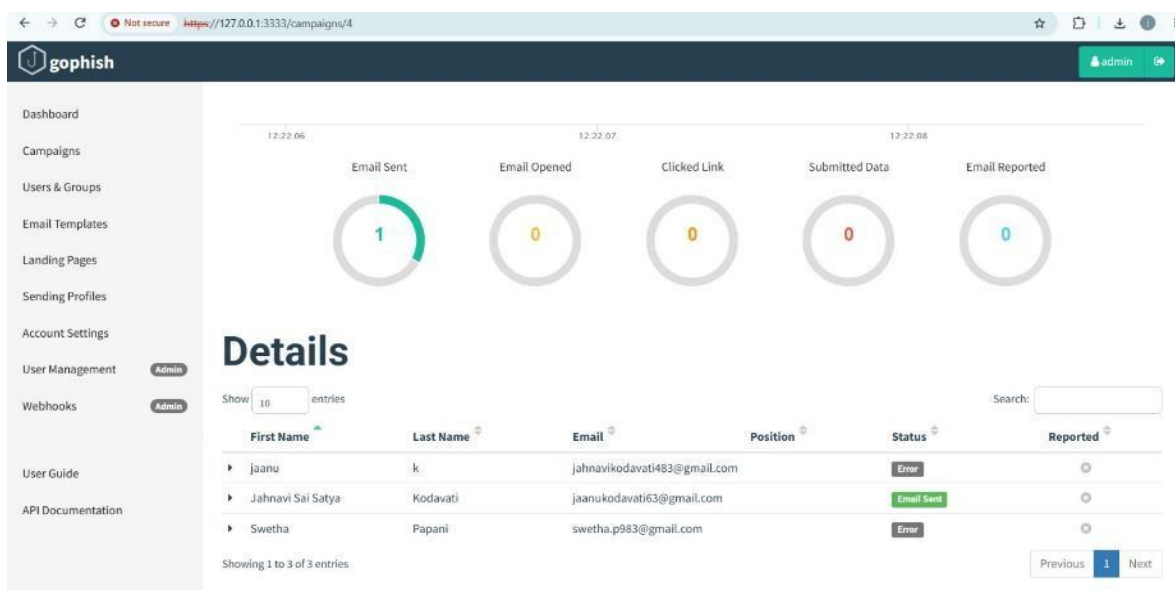
Procedure :

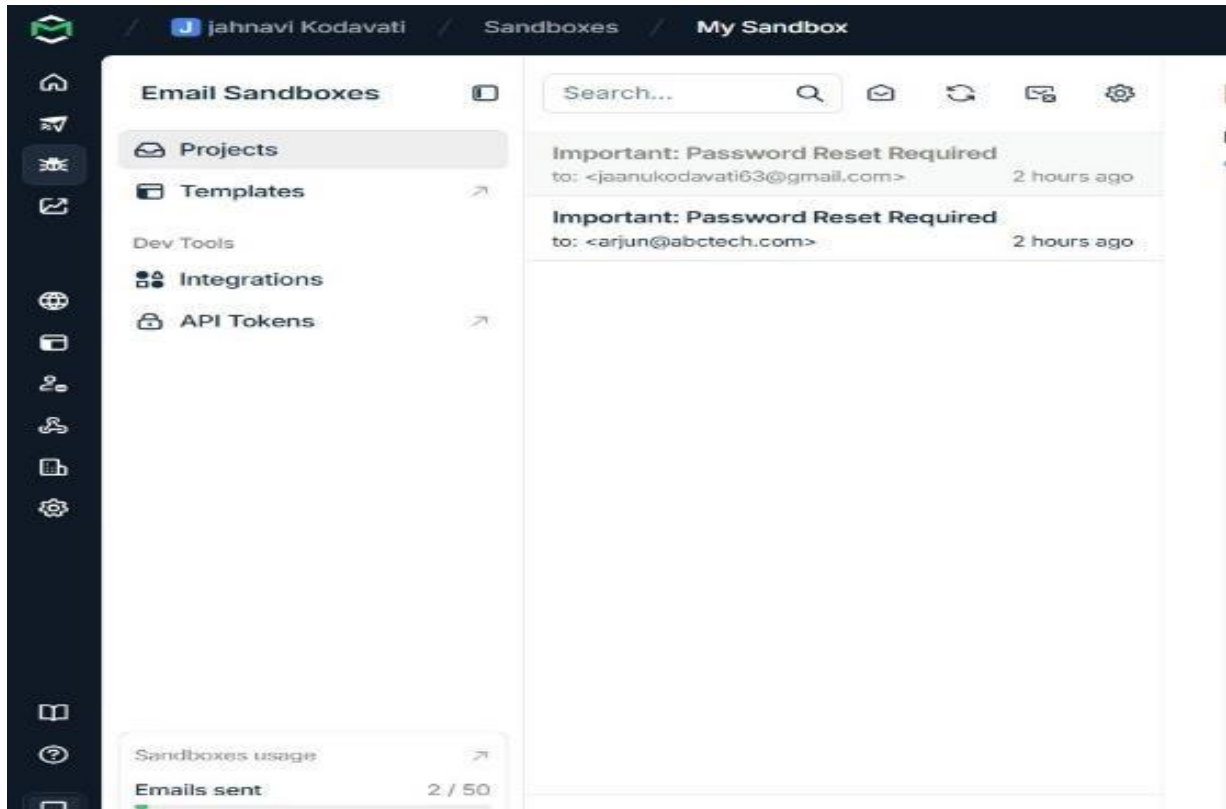
1. Created Email Template.
2. Created Landing Page.
3. Configured Mailtrap SMTP.
4. Created User Group.
5. Created Campaign.
6. Send phishing emails.
7. Viewed campaign statistics.

Metrics	Result
Target Users	3
Email Sent	1
Delivery rate	33.3%

Email Security Controls :

- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).
- Configure DKIM (DomainKeys Identified Mail).
- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).
- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering.





3. Using **Nmap Aggressive Scan (-A)** on **scanme.nmap.org**, gather detailed information about the target including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations. **(10 Marks)**

Commands :

1. `sudo nmap -A scanme.nmap.org`

Observation :

- The host is alive.
- HTTP (80) is open.
- SSH (22) is open.
- SIP ports 5060 and 5061 are open.
- Operating system information identified.

Security Recommendations :

- Keep all software updated.
- Enable firewall protection.
- Use strong authentication mechanisms.
- Perform periodic vulnerability assessments.

```
kali@kali: ~  
└─$ nmap -A scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 06:09 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.0035s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
NOT shown: 985 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    open  tcpwrapped  
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)  
53/tcp    closed domain  
80/tcp    open  http      Apache httpd 2.4.7 ((Ubuntu))  
|_http-server-header: Apache/2.4.7 (Ubuntu)  
53/tcp    closed mit-ml-dev  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
993/tcp   closed imaps  
3389/tcp  closed ms-wbt-server  
5002/tcp  closed rfe  
5060/tcp  open  sip  
5061/tcp  open  sip-tls  
8080/tcp  closed http-proxy  
8081/tcp  closed blackice-icecap  
8085/tcp  closed unknown  
Device type: VoIP adapter/bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu  
Aggressive OS guesses: AT&T BOM210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode network gateway (86%)  
No exact OS matches for host (test conditions non-ideal).  
Network Distance: 1 hop  
  
TRACEROUTE (using port 80/tcp)  
HOP RTT      ADDRESS  
1  2.76 ms scanme.nmap.org (45.33.32.156)  
  
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 123.74 seconds  
  
kali@kali: ~  
└─$
```

Code of Conduct:

*I V. Bala Siva Prasad Reddy (192424316)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

V. Bala Siva Prasad Reddy

RUBRICS FOR CALCULATION

Social Engineering Simulation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Social Engineering Scenario	20%	Demonstrates a comprehensive understanding of the social engineering scenario, attack techniques, and security context	Demonstrates good understanding with minor omissions.	Shows partial understanding but misses important aspects of the scenario.	Fails to understand or incorrectly interprets the social engineering scenario.
Identification of Attack and Vulnerabilities	20%	Correctly identifies the attack type, exploited vulnerabilities, and affected assets with complete accuracy	Correctly identifies most attack techniques with minor errors.	Identifies only some attack methods and vulnerabilities.	Unable to identify the attack type or associated vulnerabilities.
Application of Security Concepts and Countermeasures	25%	Effectively applies cybersecurity concepts and recommends appropriate preventive, detective, and corrective countermeasures.	Applies relevant concepts with minor errors and recommends suitable countermeasures.	Applies concepts partially with limited recommendations.	Unable to apply relevant cybersecurity concepts or recommend suitable countermeasures.
Risk Analysis and Decision Making	20%	Performs logical risk analysis, evaluates business impact, and provides well-justified security decisions.	Provides reasonable analysis with adequate justification.	Performs limited analysis with weak justification.	Provides no meaningful analysis or justification.
Clarity, Documentation and Technical Presentation	15%	Response is well-organized, technically accurate, professionally documented, and logically presented.	Response is clear with minor documentation or presentation issues.	Response lacks organization and contains technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.